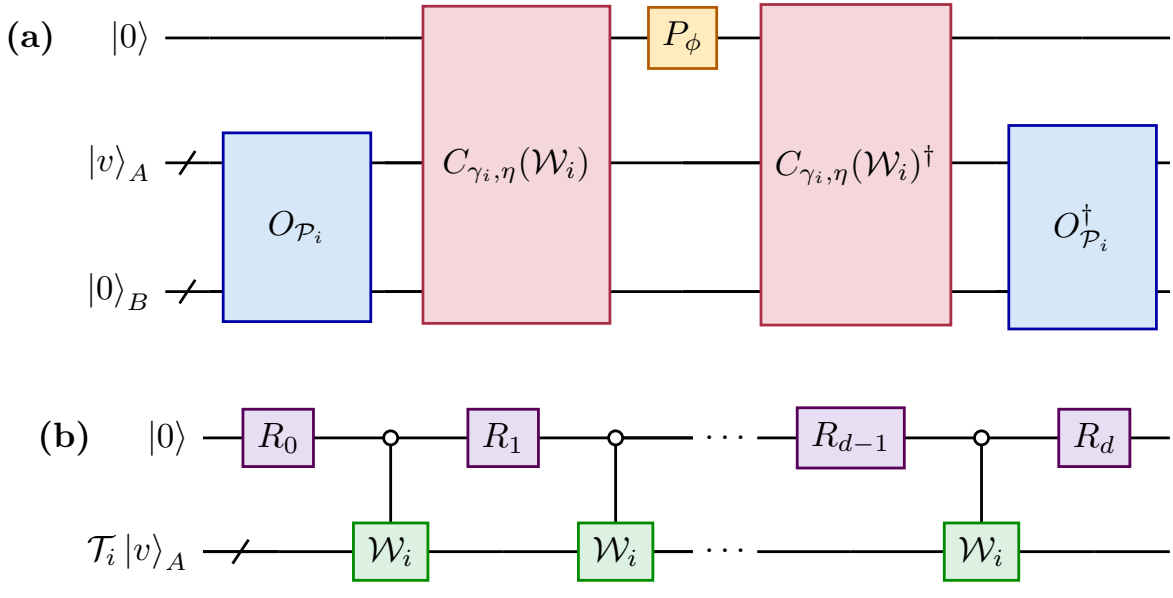

(a)
$|0\rangle$
$|v\rangle_A$
$|0\rangle_B$
$O_{\mathcal{P}_i}$
$C_{\gamma_i,\eta}(\mathcal{W}_i)$
P_ϕ
$C_{\gamma_i,\eta}(\mathcal{W}_i)^\dagger$
$O^\dagger_{\mathcal{P}_i}$
(b)
$|0\rangle$
$\mathcal{T}_i\,|v\rangle_A$
R_0
R_1
R_{d-1}
R_d
$\mathcal{W}_i$
$\mathcal{W}_i$
$\mathcal{W}_i$